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Comprehensive Analysis of Frequency-Dependent Human Body Interaction: Whole-Body Vibration, Ocular Biomechanics, and Visual Performance

1. Introduction: The Frequency Spectrum of the Human Biome

The interaction between mechanical vibration and the human physiological system represents a complex convergence of structural biomechanics, sensory processing dynamics, and neuro-control feedback loops. When the human body is subjected to Whole-Body Vibration (WBV), it ceases to function as a unified biological entity and instead behaves as a highly non-linear system of coupled masses, springs, and dampers. Each anatomical segment—from the macroscopic spinal column to the microscopic trabecular meshwork of the eye—possesses distinct natural frequencies, transmissibility characteristics, and failure modes. The study of these interactions is not merely an exercise in occupational safety compliance; it is an investigation into the fundamental operational limits of the human organism in dynamic environments.

This report provides an exhaustive, expert-level synthesis of human body frequency interactions, specifically analyzing the degradation of visual performance under vibration. By integrating data from foundational studies on vertical vibration ¹, seat-to-head transmissibility ², and advanced ocular resonance models ³, we construct a high-fidelity "frequency map" of the human body. This analysis traverses the spectrum from the low-frequency whole-body resonances (4–8 Hz) that dominate spinal dynamics, through the mid-frequency head-neck transmission (20–30 Hz), to the high-frequency resonant modes of the ocular globe (60–90 Hz).

1.1 The Theoretical Framework of Biodynamic Interaction

The human response to vibration is strictly frequency-dependent and can be categorized into three distinct regimes of interaction: the low-frequency regime dominated by gross body movement and vestibular reflexes; the mid-frequency regime governed by head-neck decoupling and resonance; and the high-frequency regime characterized by tissue-specific

resonances and intraocular dynamics.

The governing principle of this interaction is **Mechanical Impedance**, a complex function describing the resistance of the body structure to velocity forces. Defined as the ratio of driving force ($F(f)$) to the resulting velocity ($v(f)$) at a given frequency (f), impedance ($Z(f)$) encapsulates the mass, stiffness, and damping properties of the body ⁵:

$$Z(f) = \frac{F(f)}{v(f)}$$

In the context of WBV, the body acts as a lumped-parameter system. At low frequencies (below 2 Hz), the body moves in phase with the excitation source, acting largely as a rigid mass. As frequency increases, distinct segments enter resonance—a state where the excitation frequency matches the natural frequency of the structure, leading to an amplification of motion known as transmissibility.⁶ The Seat-to-Head Transmissibility (STHT) function becomes the critical determinant of visual performance, as it defines the precise magnitude and phase of the vibration reaching the visual sensors.²

1.2 The Necklace Process: A System of Systems

The transmission of vibration from the seat to the retina is often described as the "Necklace Process" ⁸, a metaphor for the interconnected chain of resonant systems. Vibration energy enters through the buttocks and thighs, travels through the pelvis, ascends the spinal column, traverses the cervical spine, enters the skull, and is finally transmitted to the ocular globe and the retinal surface. At each link in this chain, the signal is modified—amplified by resonances or attenuated by damping.

Understanding visual impairment requires analyzing this entire transmission path. A vibration that is harmless to the spine may be devastating to vision if it aligns with the resonance of the eye globe. Conversely, a frequency that causes severe spinal discomfort may be easily compensated for by the Vestibulo-Ocular Reflex (VOR). This report systematically deconstructs this chain, analyzing the physics of transmissibility, the physiological countermeasures employed by the Central Nervous System (CNS), and the ultimate failure points that result in performance degradation.

2. The Low-Frequency Domain (0.1 Hz – 4 Hz): Vestibular Dominance and Motion Sickness

The lowest frequency band, extending from 0.1 Hz to 4 Hz, is characterized by the dominance of the vestibular system. While this range is below the principal somatic resonances of the body, it is the primary domain of motion sickness (kinetosis) and low-frequency postural sway.

2.1 Vestibular Mechanics and the VOR

The Vestibulo-Ocular Reflex (VOR) is the primary physiological mechanism for stabilizing images on the retina during head motion. It functions as a feed-forward control loop, generating eye rotations equal and opposite to head rotations (Gain ≈ -1). The VOR is driven by the semicircular canals (which detect angular acceleration) and the otolith organs (which detect linear acceleration and gravity).

Frequency Response: The VOR operates with high efficiency in the 0.5 Hz to 4 Hz range.⁹ In this bandwidth, the phase lag between head motion and compensatory eye motion is minimal, allowing for effective image stabilization. Research indicates that the gain of the translational VOR (tVOR) is optimized for the frequencies encountered during human locomotion (0.5–2.0 Hz), decreasing conjugate retinal image slip and minimizing binocular disparities.⁹

However, the VOR is not perfect. It exhibits a high-pass characteristic for angular motion (AVOR) and a more complex response for linear motion (LVOR).¹⁰ As vibration frequency approaches 4 Hz, the latency of the reflex (typically 5–15 ms) begins to introduce a phase error. While a 10 ms delay is negligible at 0.5 Hz ($T=2000$ ms, phase shift $\approx 1.8^\circ$), it becomes increasingly significant as frequency rises.

2.2 Smooth Pursuit Limitations

Complementing the VOR is the smooth pursuit system, which allows the eyes to track a moving object. However, smooth pursuit is a low-bandwidth system, typically failing at frequencies above 1–2 Hz.¹⁰ In a vibration environment where a display or target is oscillating at frequencies > 2 Hz, the smooth pursuit system cannot maintain foveation. The observer must rely entirely on the VOR (if the head is moving) or suppression of the VOR (if the target moves with the head).

2.3 Motion Sickness and ISO 2631-1

The ISO 2631-1 standard recognizes the unique hazard of this frequency band through the W_f weighting curve, which peaks in the 0.1–0.5 Hz range.¹² This range corresponds to the integration time constants of the vestibular system that trigger nausea. While motion sickness does not directly reduce "visual acuity" in the sense of optical resolution, the associated symptoms (nausea, pallor, cold sweating) cause a catastrophic decline in sustained visual performance and cognitive processing.

3. The Principal Resonance (4 Hz – 8 Hz): The Spinal Engine

The frequency range of 4 to 8 Hz represents the most hostile biodynamic environment for the

seated human. This regime is dominated by the fundamental resonance of the upper body mass on the compressive stiffness of the lumbar spine.

3.1 The 5 Hz Whole-Body Resonance

At approximately 5 Hz, the seated human body exhibits its primary vertical resonance. The entire torso moves in phase with the vertical excitation, but with significant amplification. Experimental modal analysis has identified this mode as a vertical motion of the skeleton coupled with axial and shear deformations of the buttocks tissue.¹³

Visceral Dynamics: Crucially, this resonance also involves the abdominal viscera (liver, stomach, intestines), which have a natural frequency of 3–5 Hz.¹⁴ The viscera oscillate vigorously, acting as a secondary mass-spring system attached to the diaphragm and pelvic floor. This internal "sloshing" not only causes discomfort but alters the apparent mass of the body, creating a feedback loop that amplifies spinal motion.

3.2 Seat-to-Head Transmissibility (STHT) Peak

The Seat-to-Head Transmissibility (STHT) function defines the ratio of head acceleration to seat acceleration. Research consistently identifies the primary peak of vertical STHT in the **4 Hz to 5 Hz** range.² At this frequency, the head acceleration can exceed the seat acceleration by a factor of 1.5 to 2.0.

Magnitude Comparison:

Vibration Direction	Resonance Frequency	STHT Magnitude at Peak
Vertical (Z-axis)	4 – 5 Hz	1.5 – 2.0
Fore-and-Aft (X-axis)	3 – 5 Hz	1.2 – 1.8
Lateral (Y-axis)	2 – 4 Hz	1.3 – 1.6

This amplification is the primary driver of visual loss in this band. The head is subjected to large physical displacements, requiring the eyes to make large compensatory movements to maintain fixation.

3.3 Visual Performance Decrements (Below 12.5 Hz)

The core research¹ and associated studies¹ highlight that visual acuity decrements are profound below 12.5 Hz. In this range, the mechanism of impairment is **movement of the retinal image**.

Experimental Findings:

- **Thresholds:** A 10% decrease in visual acuity is observed at relatively low vibration magnitudes in this range.
- **ISO Deviation:** The experimental data suggest that the curve of equal visual impairment differs from the ISO 2631-1 W_k weighting curve. Specifically, the acceleration required to produce a 10% drop in acuity below 12.5 Hz was found to be distinct from the ISO predictions, with some subjects showing higher sensitivity (lower tolerance) at 6.3 Hz than the standard implies.¹
- **Mechanism:** The VOR, while active, cannot fully compensate for the sheer amplitude of the head displacement amplified by the spinal resonance. The "residual retinal slip" exceeds the threshold of 3–5 degrees/second¹⁷, causing image blur.

3.4 Non-Linearity and the Softening Effect

The body's response in this range is highly non-linear. As vibration magnitude increases, the resonance frequency decreases—a phenomenon known as the "softening effect".²

- **Data Point:** Increasing RMS acceleration from 0.5 m/s² to 1.5 m/s² shifts the primary resonance from ~5.5 Hz down to ~4.0 Hz.²
- **Implication:** This nonlinearity complicates the prediction of visual impairment. A suspension system tuned to dampen 5 Hz vibration may fail if the input intensity increases, shifting the body's resonance into a less-damped frequency band.

4. The Transition Zone (8 Hz – 20 Hz): Biomechanical Decoupling

As excitation frequency increases beyond 8 Hz, the body's major masses can no longer follow the rapid oscillations of the seat. The torso begins to isolate vibration (transmissibility < 1.0), but specific subsystems enter their own resonant modes.

4.1 Thoracic and Spinal Attenuation

Above 8 Hz, the large mass of the torso acts as a low-pass filter. The impedance of the spine prevents high-frequency seat vibration from effectively driving the chest and shoulders. Transmissibility drops significantly, often falling below 1.0 (attenuation) in the 10–20 Hz range.⁶

However, this attenuation is not uniform. The "Necklace Process" reveals that while the *vertical* drive to the head decreases, the *rotational* drive may increase due to the complex bending modes of the cervical and thoracic spine.

4.2 The "Back-Slap" Phenomenon

The presence of a backrest fundamentally alters transmission in this zone. Contact with a backrest couples the thorax directly to the vibration source, bypassing the damping provided by the lumbar spine.¹⁹ This is particularly critical for fore-and-aft (x-axis) vibration.

- **Mechanism:** Vertical seat vibration can be converted into fore-and-aft head motion through the geometry of the spine and the backrest interaction. This "cross-axis transmissibility" peaks in the 8–12 Hz range.
- **Vision Impact:** Fore-and-aft head motion changes the distance between the eye and the display. For a user reading a fixed display (e.g., a dashboard or monitor), this requires the accommodation (focus) system to oscillate at 10 Hz—a physiological impossibility, as accommodation has a latency of ~300 ms. The result is a constant "hunting" of the focus mechanism and significant visual fatigue.

4.3 Reading Performance and Font Size

Studies on reading performance indicate that while the 4–5 Hz range causes the most severe degradation due to gross motion, the 8–20 Hz range introduces "reading discomfort" related to character recognition.

- **Study Data:** Subjects reading text (Nimrod MT font, 8-point size) at a distance of 40 cm experienced significant difficulty at vibration magnitudes as low as 0.6 m/s^2 in this frequency band.²
- **Interference:** The frequency of vibration begins to interfere with the saccadic jumps used in reading (which occur 3–4 times per second). Vibration at 10–12 Hz creates a "stroboscopic" effect on the text line, disrupting the smooth planning of the next saccade.

5. The Head-Neck Complex (20 Hz – 30 Hz): The Pivot Point

As frequency enters the 20–30 Hz band, the head-neck system decouples from the torso and exhibits its own distinct resonant behavior. This is a critical window for visual performance, as it introduces angular perturbations that are far more detrimental to vision than linear translations.

5.1 Head Pitch Resonance (20–30 Hz)

The human head, weighing approximately 4–5 kg, is balanced on the cervical spine and controlled by a complex array of neck muscles (sternocleidomastoid, trapezius, splenius). This system exhibits a pitch (nodding) resonance typically between **20 Hz and 30 Hz**.¹⁵

Unlike the passive bounce of the 5 Hz resonance, this mode is heavily influenced by muscle tension and posture.

- **Muscle Tension:** Anxiety or the effort to stabilize the head can stiffen the neck muscles, potentially shifting this resonance frequency higher or increasing the amplitude due to reduced damping (if the muscles act as stiff springs rather than dampers).
- **Bite-Bar Measurements:** Research utilizing bite-bars to measure 6-DOF head motion confirms that while vertical transmission is low, the **angular acceleration** (pitch) peaks in this band.¹³

5.2 The Detriment of Angular Motion

Angular pitch of the head is devastating for vision. A pitch rotation of just 1 degree causes a vertical shift of the visual field that is uniform across the retina. Unlike translation, which affects near targets more than far ones, rotation degrades acuity regardless of viewing distance.

- **Retinal Slip Geometry:** For a rotation θ , the retinal slip S is roughly $S \approx \theta \times F$ (where F is focal length). Even small rotations at 25 Hz generate high retinal velocities.
- **VOR Phase Lag:** In the 20–30 Hz range, the VOR is pushed to its bandwidth limit. The latency of the reflex (5–10 ms) becomes a significant fraction of the wave period ($T \approx 40$ ms at 25 Hz). A 10 ms delay represents a **90-degree phase lag**. This means the compensatory eye movement may occur when the head has already reversed direction, potentially doubling the retinal error rather than correcting it.²²

5.3 The 25 Hz Reading Performance Dip

Specific studies identify a secondary peak of reduced reading performance at **25 Hz** during vertical vibration.² This correlates perfectly with the head-neck resonance.

- **Phenomenology:** Subjects describe this not as "shaking" but as "ghosting" or "doubling" of the text. The characters appear to smear vertically.
- **Cognitive Load:** The brain attempts to reconstruct the characters from the smeared input, leading to increased cognitive load and rapid onset of fatigue. This is a distinct failure mode from the motion-induced blur of the 5 Hz range.

6. The High-Frequency Domain (30 Hz – 60 Hz): The Gamma Band and Neural Interference

Above 30 Hz, the mechanical transmission to the head is generally low, but specific pathways (e.g., rigid headrests, helmets, high-intensity acoustics) can bridge the gap. In this domain, we must consider not just biomechanics, but the potential intersection with neurophysiology.

6.1 Neural Gamma Oscillations (30–80 Hz)

The frequency band of 30–80 Hz, known as the Gamma band, is associated with active cognitive processing, attention, and feature binding in the visual cortex.²⁴ Neural gamma oscillations are integral to the "binding problem"—the process by which the brain combines distinct visual features (color, shape, motion) into a coherent percept.

Theoretical Interference: While direct mechanical-to-neural entrainment is a subject of ongoing research, the overlap is notable. If the visual signal reaching the cortex is modulated at 40–60 Hz (due to ocular vibration), it enters the processing stream at the same frequency used for temporal binding. This presents a potential pathway for "neural dazzle" or cognitive fatigue, where the brain struggles to distinguish external vibration noise from internal processing rhythms.

6.2 The Velocity-Dependent Vision Decrement

The core study¹ notes that between 12.5 Hz and 31.5 Hz, visual acuity impairment depends primarily on vibration **velocity** (slope of 6 dB/octave). This implies that the limiting factor is the speed of the retinal image slip.

- **Integration Time:** The eye's photoreceptors integrate light over a period of 20–100 ms. At 30–60 Hz, the image traverses multiple receptors within a single integration period. This results in a loss of **contrast sensitivity**. High-spatial-frequency targets (fine details) are lost first, washed out by the blur.
- **Perceptual Shift:** The vibration is no longer perceived as "motion" but as a reduction in image quality, similar to a defocus error.

7. Ocular Resonance (60 Hz – 90 Hz): The Vibrating Shell

The most critical finding for high-frequency vision is the existence of a mechanical resonance within the eye itself. Above 60 Hz, the eye globe does not move as a rigid body; it deforms.

7.1 The Corneoscleral Shell Mode

The eye is a fluid-filled sphere (vitreous and aqueous humor) encased in a viscoelastic membrane (cornea and sclera). Like any pressurized shell, it has natural modes of vibration. Multiple sources identify the fundamental breathing mode or "bell mode" of the eye in the range of **60 Hz to 90 Hz**.¹⁴

Mechanism of Deformation:

- **Mode Shapes:** At resonance, the cornea and sclera oscillate. The cornea may bulge and flatten (piston mode) or distort asymmetrically.

- **Optical Path Distortion:** This deformation changes the axial length of the eye and the refractive power of the cornea dynamically. The distance from the lens to the retina (≈ 24 mm) oscillates, throwing the image in and out of focus 60 to 90 times per second.
- **Lens Vibration:** The crystalline lens, suspended by the zonular fibers, also has resonant modes. While its fundamental frequency is lower (15–20 Hz)²⁸, it can couple with the shell modes at higher frequencies, leading to complex internal oscillations.

7.2 Intraocular Pressure (IOP) Dependence

A profound biomechanical finding is the dependence of this resonance on Intraocular Pressure (IOP).

- **Stiffness Modulation:** As IOP increases, the "pre-stress" on the scleral shell increases, effectively stiffening the system. According to shell vibration theory, increasing stiffness raises the resonant frequency.
- **Empirical Data:** Models and experiments show that a shift in IOP from normal (~ 15 mmHg) to high (>30 mmHg) can shift the resonance frequency significantly. A 40 mmHg increase was observed to shift the resonance by **12–14 Hz**.²⁷
- **Clinical Implications:** This relationship validates the concept of **Acoustic Tonometry**.²⁹ By exciting the eye with sound waves or air puffs and measuring the resonant response, clinicians can infer IOP without corneal contact. This confirms that the 60–90 Hz band represents a true, distinct mechanical property of the ocular structure.

7.3 ISO 2631-1 and the "Missing" Band

The core article¹ highlights a critical discrepancy: the ISO 2631-1 W_k curve attenuates sensitivity rapidly above 20 Hz, effectively assuming that high-frequency vibration is benign. However, the vision data shows that sensitivity remains high or plateaus in the 31.5–63 Hz range.

- **The "Eye Resonance" Effect:** The continued visual impairment in the 40–80 Hz region is attributed to this ocular resonance. Even if the head is stabilized, if the vibration frequency excites the eye globe, vision degrades. The ISO standard, focused on spinal health and general comfort, fails to account for this organ-specific failure mode.

8. Mechanisms of Vision Loss: A Synthesis

To fully understand the data, we must synthesize the biomechanics with the sensory physiology. The degradation of vision is not a single phenomenon but a cascade of failures.

8.1 Retinal Slip Velocity Thresholds

Visual acuity is strictly limited by the velocity of the image across the retina.

- **The 3 Degree/Second Limit:** Research indicates that retinal slip velocities exceeding **3–5 degrees/second** result in a sharp decline in acuity.¹⁷
- **Calculation:** At 5 Hz, a head pitch of just 2 degrees creates a peak velocity of ~ 63 deg/s. The VOR must compensate for nearly 95% of this velocity to keep the residual slip below the 3 deg/s threshold. Any drop in VOR gain (e.g., to 0.9) results in slip velocities of >6 deg/s, causing immediate blur.

8.2 VOR Bandwidth Failure

The failure of the VOR at mid-to-high frequencies is a primary driver of vision loss.

- **Latency-Induced Phase Lag:** The 5–15 ms latency of the VOR is hard-wired.
 - At **2 Hz**: Phase lag $\approx 7^\circ$ (Negligible).
 - At **10 Hz**: Phase lag $\approx 36^\circ$ (Significant).
 - At **25 Hz**: Phase lag $\approx 90^\circ$ (Catastrophic).
- **Constructive Interference:** At 90-degree phase lag, the eye movement no longer stabilizes the image; in some parts of the cycle, it may add to the head velocity, increasing retinal slip rather than reducing it.

8.3 Intraocular "Blur" vs. Motion

At high frequencies (>40 Hz), the mechanism shifts from "motion smear" to "optical blur."

- **Refractive Oscillation:** The oscillation of the corneal curvature at 60–90 Hz introduces a dynamic astigmatism and defocus.
- **Contrast Reduction:** This does not displace the image laterally but reduces the Modulation Transfer Function (MTF) of the eye. The user perceives this as a loss of contrast and sharpness, making it difficult to resolve fine details (high spatial frequencies) even if the object appears "still."

9. Standardization and Safety: The Gap in ISO 2631-1

The International Standard ISO 2631-1 (1997) is the global benchmark for evaluating human exposure to WBV. However, the analysis in this report highlights significant discrepancies between the standard's weightings and the reality of visual performance.

9.1 The W_k Weighting Curve Critique

For vertical (z-axis) vibration, ISO 2631-1 uses the W_k frequency weighting curve.¹²

- **Characteristics:** The W_k curve is flat (highest sensitivity) between roughly 4 Hz and 12.5 Hz, reflecting the spinal resonance.
- **Vision Data Mismatch:** The core study¹ and supporting data¹⁶ show two key deviations:
 1. **Low Frequency (< 12.5 Hz):** The experimental contour for vision loss differs from the health-based W_k curve. In some cases, sensitivity at 6.3 Hz is higher than

predicted.

2. **High Frequency (> 31.5 Hz):** The W_k curve drops off steeply (attenuation), implying low risk. The vision data, driven by ocular resonance, shows sustained sensitivity. The standard effectively "ignores" the 40–80 Hz range, which is precisely where the eye globe resonates and vision blurs.

9.2 Recommendations for a "Vision Weighting"

To accurately predict visual performance in vibration environments (e.g., helicopter pilots, heavy machinery operators), a modified weighting curve is required.

- **Proposed Characteristics:**
 - **Peak:** Retain the 4–8 Hz peak (Spinal/Head transmission).
 - **Secondary Peak:** Introduce a secondary weighting peak at 20–30 Hz (Head-Neck Resonance).
 - **High-Frequency Plateau:** Remove the steep roll-off above 31.5 Hz; maintain a plateau up to 80 Hz to account for ocular shell resonance and VOR failure.

10. Conclusion: The Frequency-Defined Limits of Human Vision

The synthesis of the "Influence of whole body vibration on vision performance" paper with the broader body of biomechanical literature reveals that the human body is not a broadband absorber of vibration, but a highly tuned spectral analyzer.

Summary of Key Frequency Interactions:

Frequency Band	Dominant Mechanism	Critical Structure	Vision Impact
0.1 – 2 Hz	Vestibular Reflexes	Semicircular Canals	Motion Sickness (Nausea)
4 – 8 Hz	Whole-Body Resonance	Spine / Viscera	Severe (Retinal Displacement)
8 – 12 Hz	Thoracic Transition	Chest / Backrest	Moderate (Accommodation Hunting)

20 – 30 Hz	Head-Neck Resonance	Cervical Muscles	Severe (Ghosting / Pitch)
30 – 60 Hz	Neural / High-Freq	CNS / Gamma Band	Contrast Loss / Cognitive Load
60 – 90 Hz	Intraocular Resonance	Corneoscleral Shell	Severe (Optical Blur / Refraction)

The "Necklace Process" amplifies specific frequencies at each stage of transmission, culminating in a complex vibration spectrum at the retina. The standard ISO 2631-1, while effective for spinal health, is insufficient for predicting visual performance, particularly in the high-frequency domain where ocular resonances dominate. Future design of vehicles, displays, and diagnostic tools must account for these specific resonant bands, treating the human operator not just as a mass to be cushioned, but as a visual system with distinct, frequency-dependent failure modes.

(End of Report)

*Citations:*¹

Works cited

1. Influence of whole body vibration on vision performance - ResearchGate, accessed December 23, 2025, https://www.researchgate.net/publication/236117962_Influence_of_whole_body_vibration_on_vision_performance
2. Seat-to-Head Transmissibility and Reading Discomfort of ... - SciSpace, accessed December 23, 2025, <https://scispace.com/pdf/seat-to-head-transmissibility-and-reading-discomfort-of-the-29hne8y959.pdf>
3. Detection of Intraocular Pressure Change in a Human Eye Model Using Sonoelastic Doppler Ultrasound - University of Rochester, accessed December 23, 2025, <https://www.hajim.rochester.edu/ece/sites/parker/assets/pdf/45A%20-%20Detection%20of%20intraocular%20pressure%20change%20in%20a%20human%20eye%20model%20using%20sonoelastic%20Doppler%20Ultrasound.pdf>
4. A mechanical model of ocular bulb vibrations and implications for acoustic tonometry, accessed December 23, 2025, <https://www.medrxiv.org/content/10.1101/2023.11.10.23298373v1.full-text>
5. Theoretical relationship between vibration transmissibility and driving-point response functions of the human body - PMC - NIH, accessed December 23,

- 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4671508/>
6. The Mechanical Impedance of the Human Body in Sitting and Standing Position at Low Frequencies, accessed December 23, 2025, https://stillpointx.wordpress.com/wp-content/uploads/2015/11/the_mechanical_impedance_of_the_human_body_in_sitting_and_standing_position_at_low_frequencies.pdf
 7. Seat-to-Head Transmissibility Responses of Seated Human Body Coupled with Visco-Elastic Seats - MDPI, accessed December 23, 2025, <https://www.mdpi.com/2571-631X/5/4/51>
 8. The Necklace Process | Request PDF - ResearchGate, accessed December 23, 2025, https://www.researchgate.net/publication/243104769_The_Necklace_Process
 9. Somatosensory Influence on Platform-Induced Translational Vestibulo-Ocular Reflex in Vertical Direction in Humans - Frontiers, accessed December 23, 2025, <https://www.frontiersin.org/journals/neurology/articles/10.3389/fneur.2020.00332/full>
 10. Human Vestibuloocular Reflex and Its Interactions With Vision and Fixation Distance During Linear and Angular Head Movement | Journal of Neurophysiology | American Physiological Society, accessed December 23, 2025, <https://journals.physiology.org/doi/full/10.1152/jn.1998.80.5.2391>
 11. Amplitude and Frequency Prediction in the Translational Vestibulo-ocular Reflex - PMC, accessed December 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4486054/>
 12. ISO 2631-1:1997—Human Body Exposure to Vibration - The ANSI Blog, accessed December 23, 2025, <https://blog.ansi.org/ansi/iso-2631-1-1997-whole-body-vibration/>
 13. Resonance behaviour of the seated human body and effects of posture - PubMed, accessed December 23, 2025, <https://pubmed.ncbi.nlm.nih.gov/9593207/>
 14. Resonance frequencies of human body organs | Download Table - ResearchGate, accessed December 23, 2025, https://www.researchgate.net/figure/Resonance-frequencies-of-human-body-organs_tbl1_274471590
 15. (PDF) Vision Influence on Whole-Body Human Vibration Comfort Levels - ResearchGate, accessed December 23, 2025, https://www.researchgate.net/publication/274471590_Vision_Influence_on_Whole-Body_Human_Vibration_Comfort_Levels
 16. The effects of noise on vision efficiency - ResearchGate, accessed December 23, 2025, https://www.researchgate.net/publication/21024901_The_effects_of_noise_on_vision_efficiency
 17. Modulation of high-frequency vestibuloocular reflex during visual tracking in humans, accessed December 23, 2025, <https://cris.biu.ac.il/en/publications/modulation-of-high-frequency-vestibuloocular-reflex-during-visual-5/>

18. Seat to Head Transmissibility during Exposure to Vertical Seat Vibration: Effects of Posture and Vibration Magnitude, accessed December 23, 2025, https://iav.org/ijav/content/volumes/24_2019_1848881553860721/vol_1/1108_fullpaper_1758851553860829.pdf
19. Proceedings of the First American Conference on Human Vibration - CDC, accessed December 23, 2025, <https://www.cdc.gov/niosh/docs/2006-140/pdfs/2006-140.pdf>
20. LITERATURE ON VIBRATION AND THE HUMAN BODY, accessed December 23, 2025, <https://www.donmar.com/Tech/VIBRATION-BODY.pdf>
21. E.7 Vibration - Textbook of Maritime Health, accessed December 23, 2025, <https://textbook.maritimemedicine.com/volumes/health-risks-to-seafarers/6-7-vibration.html>
22. Effects of retinal image slip on modulation of visual vestibulo-ocular reflex during near viewing - PubMed, accessed December 23, 2025, <https://pubmed.ncbi.nlm.nih.gov/15827000/>
23. Normal Performance and Expression of Learning in the Vestibulo-Ocular Reflex (VOR) at High Frequencies - PubMed Central, accessed December 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC2603174/>
24. Right Sided Lateralization of Gamma Activity of EEG in Young Healthy Males - ResearchGate, accessed December 23, 2025, https://www.researchgate.net/profile/Menizibeya-Osain-Welcome/publication/328956972_Right_Sided_Lateralization_of_Gamma_Activity_of_EEG_in_Young_Healthy_Males/links/5bed4469a6fdcc3a8dd7540d/Right-Sided-Lateralization-of-Gamma-Activity-of-EEG-in-Young-Healthy-Males.pdf?origin=scientific-contributions
25. Gamma-range synchronization of fast-spiking interneurons can enhance detection of tactile stimuli - PMC - PubMed Central, accessed December 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC4229565/>
26. Mystery of gamma wave stimulation in brain disorders - PMC - PubMed Central, accessed December 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC11657232/>
27. Resonance frequency of fluid-filled and prestressed spherical shell—A model of the human eyeball | Request PDF - ResearchGate, accessed December 23, 2025, https://www.researchgate.net/publication/301248372_Resonance_frequency_of_fluid-filled_and_prestressed_spherical_shell-A_model_of_the_human_eyeball
28. Lens Oscillations in the Human Eye. Implications for Post-Saccadic Suppression of Vision - PubMed Central, accessed December 23, 2025, <https://pmc.ncbi.nlm.nih.gov/articles/PMC3995773/>
29. Resonance frequency of fluid-filled and prestressed spherical shell-A model of the human eyeball - PubMed, accessed December 23, 2025, <https://pubmed.ncbi.nlm.nih.gov/27106326/>
30. A mechanical model of ocular bulb vibrations and implications for acoustic tonometry, accessed December 23, 2025, https://www.researchgate.net/publication/377497586_A_mechanical_model_of_ocular_bulb_vibrations_and_implications_for_acoustic_tonometry
31. Modulation of high-frequency vestibuloocular reflex during visual tracking in

- humans - PubMed, accessed December 23, 2025,
<https://pubmed.ncbi.nlm.nih.gov/7472369/>
32. Visual acuity in the presence of retinal-image motion - ResearchGate, accessed December 23, 2025,
https://www.researchgate.net/publication/22013675_Visual_acuity_in_the_presence_of_retinal-image_motion
33. ISO-2631-1-1997.pdf - iTeh Standards, accessed December 23, 2025,
<https://cdn.standards.iteh.ai/samples/7612/0d54768e6e214481a8e814b20df83641/ISO-2631-1-1997.pdf>
34. Frequency-weighting curves W_k , W_d and W_c from ISO 2631-1 (1997). - ResearchGate, accessed December 23, 2025,
https://www.researchgate.net/figure/Frequency-weighting-curves-W-k-W-d-and-W-c-from-ISO-2631-1-1997_fig1_263897149
35. A pilot study for intraocular pressure measurements based on vibroacoustic parameters, accessed December 23, 2025,
<https://pmc.ncbi.nlm.nih.gov/articles/PMC7806898/>
36. Editorial for “Assessment of Intraocular Pressure Using Three-Dimensional MR Elastography in Ophthalmologically Normal Individ - Ovid, accessed December 23, 2025,
<https://www.ovid.com/journals/jmri/pdf/10.1002/jmri.70050~editorial-for-assessment-of-intraocular-pressure-using>